12TH CHEMISTRY · CHAPTER 19 · FULL CHAPTER MCQS

Basics of Organic Chemistry

129 topic-wise MCQs from the complete chapter with answers and reasons

INTRODUCTION AND 19.1 CATENATION

1. Organic chemistry is the study of compounds containing:

(a) Oxygen (b) Sulphur
(c) Nitrogen (d) Carbon

Answer: (d) Carbon — IUPAC definition.

2. Which of the following is NOT regarded as an organic compound?

(a) KCN (b) CH_4
(c) CCl_4 (d) CH_3OH

Answer: (a) KCN — Cyanides are traditional inorganic compounds.

3. Compared with inorganic compounds, organic compounds usually have:

(a) High melting points (b) High densities
(c) Low melting and boiling points (d) High solubility in water

Answer: (c) Low melting and boiling points — Stated in the introduction.

4. The number of known carbon-containing compounds is more than:

(a) 2 million (b) 20 million
(c) 2 thousand (d) 200 thousand

Answer: (b) 20 million — Introduction.

5. The ability of self-linkage between atoms of the same element is:

(a) Isomerism (b) Polymerisation
(c) Tautomerism (d) Catenation

Answer: (d) Catenation — Definition of catenation.

6. The word “catena” means:

(a) Chain (b) Ring
(c) Bond (d) Fat

Answer: (a) Chain — It is a Latin word.

7. Carbon forms long chains and rings because of its stable:

(a) C–H bonds (b) C–C covalent bonds
(c) Ionic bonds (d) Hydrogen bonds

Answer: (b) C–C covalent bonds — Basis of catenation.

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8. In how many different ways can four carbon atoms be arranged, according to the textbook?

(a) Two (b) Six
(c) Four (d) Eight

Answer: (c) Four — Section 19.1.

9. The valency of carbon in organic compounds is:

(a) Four (b) Two
(c) Three (d) Six

Answer: (a) Four — Carbon is tetravalent.

19.2 HYDROCARBONS

10. Hydrocarbons contain:

(a) Carbon and hydrogen only (b) Carbon, hydrogen and oxygen
(c) Carbon only (d) Carbon and oxygen only

Answer: (a) Carbon and hydrogen only — Definition.

11. Open chain hydrocarbons are also called:

(a) Aromatic (b) Aliphatic
(c) Alicyclic (d) Heterocyclic

Answer: (b) Aliphatic — Also called acyclic.

12. “Aleiphatos” is a Greek word meaning:

(a) Chain (b) Ring
(c) Fat (d) Smell

Answer: (c) Fat — Fats have open chain structures.

13. Saturated aliphatic hydrocarbons are:

(a) Alkenes (b) Arenes
(c) Alkynes (d) Alkanes

Answer: (d) Alkanes — Only single bonds.

14. Ethyne is an example of an:

(a) Alkane (b) Alkyne
(c) Alkene (d) Arene

Answer: (b) Alkyne — It has a triple bond.

15. Cyclohexane is:

(a) Aromatic (b) Heterocyclic
(c) Aliphatic open chain (d) Alicyclic

Answer: (d) Alicyclic — A cycloalkane.

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16. Aromatic hydrocarbons contain:

(a) A benzene ring or resembling structure
(b) A triple bond
(c) Only single bonds
(d) An open chain

Answer: (a) A benzene ring or resembling structure — Definition.

17. Which of the following is an aromatic hydrocarbon?

(a) Cyclohexane (b) Ethene
(c) Naphthalene (d) Cyclohexene

Answer: (c) Naphthalene — It has fused benzene rings.

18. Alicyclic hydrocarbons behave like:

(a) Aromatic hydrocarbons (b) Acids
(c) Alcohols (d) Aliphatic hydrocarbons

Answer: (d) Aliphatic hydrocarbons — Hence the name.

19.3 FUNCTIONAL GROUPS

19. The characteristic properties of an organic compound are determined by its:

(a) Carbon chain (b) Molecular mass
(c) Functional group (d) Hydrogen atoms

Answer: (c) Functional group — Definition of functional group.

20. Alcohols have higher boiling points than the corresponding hydrocarbons because of:

(a) Hydrogen bonding (b) Greater mass only
(c) Ionic bonding (d) Double bonds

Answer: (a) Hydrogen bonding — The –OH group forms hydrogen bonds.

21. Which class has no functional group?

(a) Alkenes (b) Alkanes
(c) Alkynes (d) Alcohols

Answer: (b) Alkanes — No polarity and no multiple bonds.

22. Glucose and sucrose dissolve in water due to their:

(a) Large size (b) Sweet taste
(c) Carbon chains (d) Polar functional groups

Answer: (d) Polar functional groups — Section 19.3.

23. At room temperature methanol is a liquid while methane is a:

(a) Gas (b) Solid
(c) Liquid (d) Plasma

Answer: (a) Gas — Effect of the –OH group.

24. The functional group of an alkene is:

(a) A C–C single bond (b) A hydroxyl group
(c) A C=C double bond (d) A carbonyl group

Answer: (c) A C=C double bond — A multiple bond can be a functional group.

19.4 NOMENCLATURE

25. IUPAC stands for International Union of Pure and:

(a) Analytical Chemistry (b) Advanced Chemistry
(c) Applied Chemistry (d) Atomic Chemistry

Answer: (c) Applied Chemistry — Section 19.4.

26. The stem (root) of an IUPAC name gives the:

(a) Functional group (b) Longest continuous carbon chain
(c) Side chain (d) Number of double bonds

Answer: (b) Longest continuous carbon chain — Component 1.

27. The primary suffix tells whether the parent chain is:

(a) Cyclic or open (b) Long or short
(c) Polar or non-polar (d) Saturated or unsaturated

Answer: (d) Saturated or unsaturated — an, en, yn.

28. The secondary suffix gives the:

(a) Family (functional group) name (b) Chain length
(c) Position of branches (d) Number of rings

Answer: (a) Family (functional group) name — Suffix of the key functional group.

29. Side chains and substituents are shown by the:

(a) Suffix (b) Root
(c) Stem (d) Prefix

Answer: (d) Prefix — With the position number.

30. A ring of carbon atoms is shown in the name by:

(a) iso (b) neo
(c) cyclo (d) oxo

Answer: (c) cyclo — The name starts with cyclo.

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31. The IUPAC name of alkyl halides is:

(a) Halogenoalkane (b) Alkanol
(c) Alkanal (d) Alkanoyl halide

Answer: (a) Halogenoalkane — Class 1.

32. The IUPAC name of $CH_3CH(Br)CH_3$ is:

(a) 1-Bromopropane (b) 2-Bromopropane
(c) Bromoethane (d) 2-Bromobutane

Answer: (b) 2-Bromopropane — Br on C-2 of propane.

33. The IUPAC name of $(CH_3)_3COH$ is:

(a) Butan-1-ol (b) 2-Methylpropan-1-ol
(c) Butan-2-ol (d) 2-Methylpropan-2-ol

Answer: (d) 2-Methylpropan-2-ol — A tertiary alcohol.

34. The suffix for aldehydes is:

(a) -one (b) -ol
(c) -al (d) -oic acid

Answer: (c) -al — Alkanal.

35. The suffix for ketones is:

(a) -one (b) -al
(c) -ol (d) -oate

Answer: (a) -one — Alkanone.

36. CH_3COCH_3 is named:

(a) Propanal (b) Propanone
(c) Propanol (d) Propanoic acid

Answer: (b) Propanone — A three-carbon ketone.

37. The IUPAC name of CH_3COOH is:

(a) Methanoic acid (b) Propanoic acid
(c) Ethanoic acid (d) Ethanal

Answer: (c) Ethanoic acid — Two carbon atoms.

38. CH_3CONH_2 is:

(a) Ethanamide (b) Ethanamine
(c) Ethanenitrile (d) Methanamide

Answer: (a) Ethanamide — An acid amide.

39. CH_3COCl is named:

(a) Chloroethane (b) Chloroethanal
(c) Ethyl chloride (d) Ethanoyl chloride

Answer: (d) Ethanoyl chloride — An acid halide.

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40. $CH_3COOCH_2CH_3$ is:

(a) Methyl ethanoate (b) Ethyl ethanoate
(c) Ethyl methanoate (d) Methyl propanoate

Answer: (b) Ethyl ethanoate — Alkyl alkanoate.

41. The IUPAC name of ethers is:

(a) Alkanol (b) Alkyl alkanoate
(c) Alkoxyalkane (d) Alkanone

Answer: (c) Alkoxyalkane — Class 10.

42. $CH_3CH_2OCH_2CH_3$ is:

(a) Methoxyethane (b) Ethoxymethane
(c) Methoxymethane (d) Ethoxyethane

Answer: (d) Ethoxyethane — Two ethyl groups on oxygen.

43. CH_3CN is named:

(a) Ethanenitrile (b) Methanenitrile
(c) Ethanamine (d) Methanamide

Answer: (a) Ethanenitrile — The CN carbon is counted.

44. $(CH_3CO)_2O$ is:

(a) Ethyl ethanoate (b) Ethanoic anhydride
(c) Ethanoic acid (d) Ethoxyethane

Answer: (b) Ethanoic anhydride — An acid anhydride.

45. $CH_3CH(NH_2)CH_3$ is:

(a) Propan-1-amine (b) N-Methylethanamine
(c) Ethanamine (d) Propan-2-amine

Answer: (d) Propan-2-amine — NH2 on C-2.

46. An amine containing the –NH–R group is:

(a) Primary (b) Tertiary
(c) Secondary (d) Quaternary

Answer: (c) Secondary — Two alkyl groups on N.

47. $CH_3CH_2CH_2N(CH_3)_2$ is:

(a) N,N-Dimethylpropan-1-amine
(b) N-Methylpropan-1-amine
(c) Propan-2-amine
(d) N-Ethylpropan-1-amine

Answer: (a) N,N-Dimethylpropan-1-amine — A tertiary amine.

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48. The functional group of nitroalkanes is:

(a) $-NH_2$
(b) $-NO_2$
(c) –CN
(d) $-CONH_2$

Answer: (b) $-NO_2$ — Class 13.

49. The general formula $C_nH_{2n}O$ belongs to:

(a) Alcohols and ethers
(b) Carboxylic acids and esters
(c) Aldehydes and ketones
(d) Amines

Answer: (c) Aldehydes and ketones — Table of classes.

50. Carboxylic acids and esters share the general formula:

(a) $C_nH_{2n}O$
(b) $C_nH_{2n}O_2$
(c) $C_nH_{2n+2}O$
(d) $C_nH_{2n+1}X$

Answer: (b) $C_nH_{2n}O_2$ — Table of classes.

51. The systematic name of CH_3CCl_3 is:

(a) 1,1,1-Trichloroethane
(b) 1,1,2-Trichloroethane
(c) Trichloromethane
(d) 1,2-Dichloroethane

Answer: (a) 1,1,1-Trichloroethane — All three Cl on one carbon.

19.5 AND 19.6 REPRESENTING MOLECULES; MOLECULAR FORMULA

52. The formula showing the actual number of atoms of each element in a molecule is the:

(a) Molecular formula
(b) Empirical formula
(c) Skeletal formula
(d) Displayed formula

Answer: (a) Molecular formula — Definition.

53. The empirical formula of butane is:

(a) C_4H_{10}
(b) CH_3
(c) CH_2
(d) C_2H_5

Answer: (d) C_2H_5 — Simplest ratio of C4H10.

54. The empirical formula of ethanoic acid is:

(a) $C_2H_4O_2$
(b) CHO
(c) CH_2O
(d) CH_3O

Answer: (c) CH_2O — Simplest ratio.

55. Ethanol and methoxymethane both have the molecular formula:

(a) C_2H_4O (b) C_2H_6O
(c) C_3H_8O (d) CH_4O

Answer: (b) C_2H_6O — They are isomers.

56. The formula that shows all the bonds between the atoms is the:

(a) Condensed formula (b) Empirical formula
(c) Displayed formula (d) Molecular formula

Answer: (c) Displayed formula — Also called graphical formula.

57. A one-line structure in which single bonds are omitted is the:

(a) Condensed formula (b) Skeletal formula
(c) Displayed formula (d) Empirical formula

Answer: (a) Condensed formula — Also called linear formula.

58. The condensed formula $CH_3(CH_2)_2CH_3$ represents:

(a) Propane (b) Butane
(c) Pentane (d) Ethane

Answer: (b) Butane — Four carbon atoms.

59. In a skeletal formula, each joint or line end represents a:

(a) Hydrogen atom (b) Lone pair
(c) Functional group (d) Carbon atom

Answer: (d) Carbon atom — Unless another atom is shown.

60. In a skeletal formula, a double line represents a:

(a) Ring (b) Double bond
(c) Lone pair (d) Hydrogen bond

Answer: (b) Double bond — Rule 2.

61. Skeletal formulae are also called:

(a) Bond-line structures (b) Dot-cross structures
(c) Graphical formulae (d) Linear formulae

Answer: (a) Bond-line structures — Section 19.5.

62. The molecular formula of cyclohexane is:

(a) C_6H_6 (b) C_6H_{14}
(c) C_6H_{12} (d) C_6H_{10}

Answer: (c) C_6H_{12} — Six CH2 groups.

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63. The molecular formula of ethanamide is:

(a) C_2H_7N
(b) C_2H_3N
(c) CH_3NO
(d) C_2H_5NO

Answer: (d) C_2H_5NO — CH_3CONH_2.

19.7 UNDERSTANDING ORGANIC REACTIONS

64. Homolytic fission occurs when the breaking bond is:

(a) Non-polar (b) Polar
(c) Ionic (d) Coordinate

Answer: (a) Non-polar — Electrons are shared equally.

65. Homolytic fission produces:

(a) Ions (b) Carbocations
(c) Free radicals (d) Carbanions

Answer: (c) Free radicals — Each atom gets one electron.

66. Heterolytic fission produces:

(a) Free radicals (b) Molecules
(c) Atoms (d) Ions

Answer: (d) Ions — One atom takes both electrons.

67. A species with a positively charged carbon atom is a:

(a) Carbanion (b) Carbocation
(c) Free radical (d) Nucleophile

Answer: (b) Carbocation — Definition.

68. A carbanion is formed when carbon is bonded to:

(a) A metal such as Li or Mg (b) Chlorine
(c) Oxygen (d) Nitrogen

Answer: (a) A metal such as Li or Mg — An atom less electronegative than carbon.

69. Free radicals are highly reactive because they have:

(a) A positive charge (b) A full octet
(c) A lone pair (d) An unpaired electron

Answer: (d) An unpaired electron — It tends to pair up.

70. The species that undergoes attack in a reaction is the:

(a) Reagent (b) Catalyst
(c) Substrate (d) Product

Answer: (c) Substrate — The attacking species is the reagent.

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71. An electrophile is able to:

(a) Donate an electron pair (b) Accept an electron pair
(c) Donate a proton only (d) Lose an electron

Answer: (b) Accept an electron pair — Electron lover.

72. Which of the following is an electrophile?

(a) NH_3 (b) CN^-
(c) OH^- (d) BF_3

Answer: (d) BF_3 — A neutral electron-deficient molecule.

73. Which of the following is a nucleophile?

(a) H_2O
(b) NO_2^+
(c) SO_3
(d) H^+

Answer: (a) H_2O — It has lone pairs to donate.

74. The word nucleophile means:

(a) Electron lover (b) Nucleus lover
(c) Water lover (d) Proton hater

Answer: (b) Nucleus lover — It attacks positive centres.

75. In CH_3–Li the carbon atom is:

(a) Electrophilic (b) A free radical
(c) Nucleophilic (d) Neutral

Answer: (c) Nucleophilic — It carries δ–.

76. In CH_3–OH the carbon atom is:

(a) Nucleophilic (b) Electrophilic
(c) A free radical (d) Negatively charged

Answer: (b) Electrophilic — It carries δ+.

77. $CH_3CH_2Br + KOH_{(aq)} \rightarrow CH_3CH_2OH + KBr$ is:

(a) Addition (b) Condensation
(c) Elimination (d) Substitution

Answer: (d) Substitution — Br is replaced by OH.

78. A reaction in which two reactants give a single product is:

(a) Addition (b) Substitution
(c) Elimination (d) Hydrolysis

Answer: (a) Addition — Definition.

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79. Bromoethane with alcoholic KOH gives ethene. This is:

(a) Substitution (b) Addition
(c) Elimination (d) Oxidation

Answer: (c) Elimination — HBr is removed from adjacent carbons.

80. The catalyst for the addition of hydrogen to ethene is:

(a) Platinum oxide (b) Nickel
(c) Iron (d) Vanadium pentoxide

Answer: (b) Nickel — At 200–300 °C.

81. Breaking of a compound into two by reaction with water is:

(a) Condensation (b) Reduction
(c) Addition (d) Hydrolysis

Answer: (d) Hydrolysis — Often acid or base catalysed.

82. Acid hydrolysis of ethyl ethanoate gives ethanoic acid and:

(a) Ethanol (b) Methanol
(c) Ethanal (d) Ethene

Answer: (a) Ethanol — Section 19.7 C.

83. Propanone reacts with hydroxylamine to form an oxime. This is:

(a) Hydrolysis (b) Elimination
(c) Condensation (d) Reduction

Answer: (c) Condensation — Water is lost.

84. In organic chemistry, oxidation involves:

(a) Gain of hydrogen (b) Loss of oxygen
(c) Gain of electrons (d) Gain of oxygen or loss of hydrogen

Answer: (d) Gain of oxygen or loss of hydrogen — Number of C–O bonds increases.

85. Ozonolysis of propene gives ethanal and:

(a) Propanal (b) Methanal
(c) Ethanol (d) Propanone

Answer: (b) Methanal — O_3, then Zn/H_2O.

86. Elimination is the reverse of:

(a) Addition (b) Substitution
(c) Hydrolysis (d) Oxidation

Answer: (a) Addition — A multiple bond is formed instead of used.

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19.8 MECHANISMS OF CHEMICAL REACTIONS

87. The pathway of a reaction showing bond breaking and bond making is its:

(a) Rate (b) Mechanism
(c) Yield (d) Equilibrium

Answer: (b) Mechanism — Definition.

88. The free radical mechanism involves:

(a) Homolysis (b) Heterolysis
(c) Hydrolysis (d) Ionisation

Answer: (a) Homolysis — Free radicals are produced.

89. The stage in which the initial free radicals are produced is:

(a) Propagation (b) Substitution
(c) Termination (d) Initiation

Answer: (d) Initiation — First stage.

90. The stage in which free radicals combine with each other is:

(a) Initiation (b) Propagation
(c) Termination (d) Elimination

Answer: (c) Termination — Last stage.

91. Methane reacts with chlorine in diffused sunlight to finally give:

(a) CH_3Cl only (b) $CHCl_3$ only
(c) C_2H_6 (d) CCl_4

Answer: (d) CCl_4 — All four H atoms are substituted.

92. The ionic or polar mechanism involves:

(a) Heterolysis (b) Homolysis
(c) Free radicals (d) Photolysis only

Answer: (a) Heterolysis — Electrophiles and nucleophiles are produced.

93. The nucleophile that is replaced in nucleophilic substitution is the:

(a) Substrate (b) Leaving group
(c) Electrophile (d) Catalyst

Answer: (b) Leaving group — Section 19.8.

94. Carbonyl compounds typically undergo:

(a) Electrophilic addition (b) Free radical substitution
(c) Nucleophilic addition (d) Electrophilic substitution

Answer: (c) Nucleophilic addition — The carbonyl carbon is electrophilic.

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95. Alkenes typically undergo:

(a) Nucleophilic addition (b) Hydrolysis
(c) Nucleophilic substitution (d) Electrophilic addition

Answer: (d) Electrophilic addition — The double bond is electron-rich.

96. In elimination, two atoms or groups are removed from:

(a) Two adjacent carbons (b) The same carbon
(c) Two distant carbons (d) A ring only

Answer: (a) Two adjacent carbons — A multiple bond forms.

19.9 ISOMERISM

97. Isomers have the same:

(a) Molecular formula (b) Structure
(c) Physical properties (d) Chemical properties

Answer: (a) Molecular formula — But different arrangement of atoms.

98. Isomers with different bond connectivity are:

(a) Stereoisomers (b) Enantiomers
(c) Constitutional isomers (d) Conformers

Answer: (c) Constitutional isomers — Structural isomers.

99. Which is NOT a type of structural isomerism?

(a) Metamerism (b) Chain isomerism
(c) Tautomerism (d) Optical isomerism

Answer: (d) Optical isomerism — It is stereoisomerism.

100. Stereoisomers differ in:

(a) Molecular formula (b) Spatial arrangement of groups
(c) Bond connectivity (d) Number of atoms

Answer: (b) Spatial arrangement of groups — Different configuration.

101. Optical activity is measured with a:

(a) Polarimeter (b) Spectrometer
(c) Calorimeter (d) Colorimeter

Answer: (a) Polarimeter — Using plane polarised light.

102. Plane polarised light vibrates in:

(a) All planes (b) Two planes
(c) Only one plane (d) No plane

Answer: (c) Only one plane — Monochromatic light of single wavelength.

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103. A compound that rotates plane polarised light clockwise is:

(a) Levorotatory (b) Meso
(c) Racemic (d) Dextrorotatory

Answer: (d) Dextrorotatory — Shown by (+).

104. A levorotatory compound is shown by the sign:

(a) (+) (b) (–)
(c) (±) (d) (d)

Answer: (b) (–) — Anticlockwise rotation.

105. A carbon atom bonded to four different groups is:

(a) Achiral (b) Unsaturated
(c) Planar (d) Chiral

Answer: (d) Chiral — Chiral centre.

106. The word "chiral" comes from the Greek word for:

(a) Hand (b) Mirror
(c) Light (d) Opposite

Answer: (a) Hand — Cheir; handedness.

107. A molecule that is superimposable on its mirror image is:

(a) Chiral (b) Dextrorotatory
(c) Achiral (d) An enantiomer

Answer: (c) Achiral — Optically inactive.

108. The number of chiral carbon atoms in lactic acid is:

(a) Two (b) One
(c) Three (d) Zero

Answer: (b) One — Bonded to CH_3, H, OH and COOH.

109. Non-superimposable mirror image isomers are:

(a) Enantiomers (b) Diastereomers
(c) Conformers (d) Metamers

Answer: (a) Enantiomers — "Enantio" means opposite.

110. Enantiomers differ in:

(a) Boiling point
(b) Density
(c) Direction of rotation of plane polarised light
(d) Melting point

Answer: (c) Direction of rotation of plane polarised light — Other properties are identical.

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111. A racemic mixture is shown by:

(a) (+) (b) (m)
(c) (–) (d) (±)

Answer: (d) (±) — 50% (+) and 50% (–).

112. A racemic mixture is:

(a) Dextrorotatory (b) Optically inactive
(c) Levorotatory (d) A meso compound

Answer: (b) Optically inactive — The rotations cancel.

113. A meso form is optically inactive due to:

(a) A plane of symmetry (b) Absence of chiral carbons
(c) A double bond (d) Free rotation

Answer: (a) A plane of symmetry — Internal compensation.

114. The classic example of a meso compound is:

(a) Lactic acid (b) Thalidomide
(c) But-2-ene (d) Tartaric acid

Answer: (d) Tartaric acid — Two similar chiral carbons.

115. The maximum number of optical isomers for n chiral carbons is:

(a) 2n (b) n^2
(c) 2^n (d) n + 2

Answer: (c) 2^n — The 2^n rule.

116. A compound with two different chiral carbons has how many optical isomers?

(a) Two (b) Four
(c) Three (d) Eight

Answer: (b) Four — $2^2 = 4$.

117. Cis-trans isomerism arises from:

(a) Restricted rotation about a double bond
(b) Free rotation about a single bond
(c) A chiral carbon
(d) Hydrogen bonding

Answer: (a) Restricted rotation about a double bond — Definition.

118. The doubly bonded carbons of but-2-ene are:

(a) sp hybridized (b) sp^3 hybridized
(c) sp^2 hybridized (d) Unhybridized

Answer: (c) sp^2 hybridized — One σ and one π bond.

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119. The energy needed to break the π bond for rotation is:

(a) 163 kJ/mol (b) 263 kJ/mol
(c) 363 kJ/mol (d) 63 kJ/mol

Answer: (b) 263 kJ/mol — Not available at room temperature.

120. Forms of butane obtained by rotation about the C–C bond are:

(a) Cis-trans isomers (b) Position isomers
(c) Enantiomers (d) Conformational isomers

Answer: (d) Conformational isomers — They cannot be separated at room temperature.

121. In the cis isomer the similar groups are:

(a) On opposite sides (b) On the same carbon
(c) On the same side (d) Absent

Answer: (c) On the same side — Trans means opposite sides.

122. Which compound does NOT show cis-trans isomerism?

(a) But-2-ene (b) But-1-ene
(c) Pent-2-ene (d) 1,2-Dichloroethene

Answer: (b) But-1-ene — C-1 has two identical H atoms.

123. Compared with cis isomers, trans isomers are generally:

(a) More stable (b) Less stable
(c) Equally stable (d) Optically active

Answer: (a) More stable — Section 19.9.

19.10 STEREOCHEMISTRY IN DRUG DEVELOPMENT

124. Thalidomide was used as a:

(a) Painkiller (b) Antibiotic
(c) Tranquilliser (sedative) (d) Antiseptic

Answer: (c) Tranquilliser (sedative) — Also given for morning sickness.

125. Thalidomide was originally distributed as:

(a) The pure (+) isomer (b) A racemic mixture
(c) The pure (–) isomer (d) A meso form

Answer: (b) A racemic mixture — Section 19.10.

126. Which isomer of thalidomide damaged the fetus?

(a) (+) isomer (b) Neither
(c) Both equally (d) (–) isomer

Answer: (d) (–) isomer — The (+) isomer was a safe tranquilliser.

127. Pure (+) thalidomide is still unsafe because in the blood it:

(a) Racemises rapidly (b) Decomposes
(c) Precipitates (d) Is oxidized

Answer: (a) Racemises rapidly — Over 50% in 10 minutes.

128. Drug receptors in the body act as:

(a) Catalysts (b) Solvents
(c) Chiral locks (d) Racemic mixtures

Answer: (c) Chiral locks — They are highly specific.

129. The thalidomide tragedy was discovered in the early:

(a) 1940s (b) 1920s
(c) 1980s (d) 1960s

Answer: (d) 1960s — The drug was withdrawn immediately.

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